

Quick Check

Lesson 5: Primary and Secondary Growth: How Do Plants Show Growth?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 5

A farmer in Kakamega watches young maize and bean seedlings after germination. Which parts of each seedling are mainly getting longer?

- A** The whole stem and root grow longer evenly, like an animal growing all over
- B** Only the bark between the xylem and phloem
- C** The tips of the shoot and the tips of the root
- D** Only the old leaves near the soil

A learner views a low-power section of a young bean stem and a young maize stem under a microscope. How are the vascular bundles arranged?

- A** Both stems have bundles in a ring
- B** Bean: scattered through the stem. Maize: in a ring
- C** Bean: in a ring. Maize: scattered through the stem
- D** Both stems have a single bundle in the centre

A learner counts 14 rings on a cut Eucalyptus log. What do the rings and the thick trunk show about the tree?

- A** The trunk has grown only in length through primary growth
- B** The trunk has increased in girth through secondary growth
- C** The trunk has thickened by stretching of existing cells only
- D** The trunk has no cambium

Bean plants thicken and branch, but maize stems stay mostly the same width. What is the best explanation?

- A** Maize stems have a cambium ring but it is too thin to work
- B** Bean stems thicken only because they take in more water
- C** Maize stems have scattered vascular bundles with no cambium, so they do not thicken much
- D** Maize does not have a shoot tip, so it cannot grow

In a 3-step model of secondary growth in a Grevillea stem, the cambium forms a complete ring. What does this ring of cambium produce?

- A** Only cork cells on both sides
- B** New root tips at the base of the stem
- C** Secondary phloem inward and secondary xylem outward
- D** Secondary xylem inward and secondary phloem outward

A learner revises a model of maize (hypogeal) and bean (epigeal) seedlings after germination. Which statement should be in the model?

- A** Both seedlings start growing in length by primary growth from the shoot and root apical meristems
- B** Maize grows only by secondary growth, and bean grows only by primary growth
- C** Neither seedling has any meristem
- D** Bean seedlings start with secondary growth, but maize seedlings start with primary growth

Which pair correctly matches the type of growth with the meristem and what increases?

- A** Primary growth: lateral meristems, girth increases
- B** Primary growth: apical meristems, length increases
- C** Secondary growth: apical meristems, length increases
- D** Secondary growth: apical meristems, girth increases

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).